

List of Projects - Summer 2026

Project No.	Project Title (Max 200 characters)	A brief writeup of the proposal (Max 150 words)	First Mentor	Contact Number	Email Address	Other Mentors	E-mail ID	Contact Number	Name of the Department
1	Upgrade of Rapid Milk Adulteration Detection Kit to Smart Sensor-Based Food Quality Monitoring System	Building upon the existing ELC activity on the development of a rapid adulteration detection kit for milk, this project proposes an advanced, technology-driven upgrade to enhance functionality, accuracy, and real-time usability. While the earlier activity focused on chemical-based detection of common adulterants (such as starch, soap, acids, and sugars), the upgraded system aims to incorporate low-cost sensor technology for continuous and real-time monitoring of food quality. The proposed system will integrate sensors (e.g., MQ-series gas sensors, pH probes, temperature, and humidity sensors) with a microcontroller platform to detect spoilage, microbial activity, and quality deterioration in dairy products. The system will analyze sensor data patterns associated with food degradation and generate a simple quality index or alert for users. This transition from manual detection kits to a smart sensing platform represents a significant improvement in usability, scalability, and practical application. The upgraded activity will provide students with interdisciplinary exposure spanning food science, sensor technology, embedded systems, and basic data analytics, thereby enhancing experiential learning outcomes. Expected Outcomes: • Development of a prototype smart food quality monitoring device • Comparison between traditional adulteration tests and sensor-based detection • Real-time monitoring and alert system for milk spoilage and quality deterioration • Improved understanding of food safety, instrumentation, and data-driven analysis • A scalable solution suitable for households and small-scale vendors Novelty / Upgrade Component: • Transition from manual chemical testing kit → smart sensor-based system • Inclusion of real-time monitoring instead of one-time testing • Integration of electronics and data analysis with food quality assessment • Expansion from adulteration detection → overall food quality & spoilage monitoring	Dr. Ravi Kumar	9501020747	ravi.kumar@thapar.edu	Dr. Ovais Shafiq Qadri	osqadri@thapar.edu	7007660275	BTD
2	Tiny Changes, Big Impact: Detecting Genetic Disorders Through SNPs	A single-nucleotide polymorphism (SNP) is a variation in the DNA sequence at a specific position among individuals, leading to genetic variation among people. For instance, an SNP might switch out the nucleotide cytosine (C) for the nucleotide thymine (T) in a certain part of DNA. Most commonly, these variations are found in the DNA between genes. When SNPs occur within a gene or in a regulatory region near it, they may play a more direct role in disease by affecting the gene's function. SNPs can also be used to track the inheritance of disease genes within families. SNPs occur naturally throughout a person's DNA and function as biological markers that help scientists detect genes associated with a particular genetic disorder. Some of these genetic differences, however, have proven crucial to the study of human health. Although a particular SNP may not cause a disorder, some SNPs are associated with certain diseases. Molecular characterization relies on the detection of these diseases, and SNP-based techniques facilitate diagnosis, prevention, and treatment.	Dr. Siddharth Sharma	9501688366	siddharthsharma.phd@thapar.edu	Dr. Atul Upadhyay	siddharthsharma.phd@thapar.edu	8088810529	BTD

3	Structural Health Monitoring & Development of Maintenance Management System for Rigid Pavements using AI/ML	The proposed study aims to develop an AI-based distress detection and classification system for pavement infrastructure using advanced deep learning techniques. The system will be capable of automatically identifying different types of pavement distress, such as cracks, potholes, rutting, ravelling, and surface deformation, from field images or video data. In addition to detecting the distress, the model will localize the damaged regions and assess the severity level with high accuracy and reliability. Such an approach can significantly reduce manual inspection efforts, improve consistency in condition assessment, and enable rapid monitoring of large pavement networks. The study also proposes the development of an automated maintenance scheduling and alert system integrated with the distress detection framework. The maintenance recommendations will be generated in accordance with the provisions of IS 456:2000 and IRC 83:2018. By incorporating historical pavement performance data, real-time distress information, and AI-based predictions, the system will support optimized maintenance planning, resource allocation, and timely intervention strategies. A mobile application interface will further facilitate field-level implementation by enabling engineers and maintenance personnel to receive alerts, review distress severity, and plan repair activities efficiently.	Dr. Himanshu Chawla	8233114245	himanshu.chawla@thapar.edu	Dr. Tanuj Chopra, Dr. Amit Trivedi & Dr. Shruti Sharma	tchopra@thapar.edu	7837122924	CED
4	Stability of earthen dam	Experiments will be carried out to assess the stability of the earthen dam using different types of soil materials. These will be conducted with a physical model, and the results will be validated through numerical models. The experiments will be expanded to improve stability using various low-cost and nature-based soil reinforcements, and the most effective method will be reported.	Dr. Tapas Karmaker	8054581075	tkarmaker@thapar.edu	Dr. Richa Babbar	richa.babbar@thapar.edu	9780310043	CED
5	Set for Essential Oil Extraction from Turmeric Rhizomes using Solar Energy	Oil Extraction project to be continued	Dr. Avinash Chandra	9914255295	avinash.chandra@thapar.edu	Dr. Ravinder Kumar Duvedi	rduvedi@thapar.edu	9780474711	CHED
6	AI-Augmented High-Performance Modeling of Heat and Mass Transfer in Fluid Systems.	This study proposes a hybrid computational framework combining artificial intelligence and numerical methods for modeling heat and mass transfer in fluid systems. Classical governing equations, including fluid flow and energy equations, are solved using scalable solvers from the Portable, Extensible Toolkit for Scientific Computation. To enhance efficiency and capture complex nonlinear dynamics, convolutional neural networks (CNNs) are integrated for spatial feature extraction, while neural network-based PDE solvers are employed to approximate solution behavior. A physics-informed learning strategy is adopted to enforce governing equations within the training process, ensuring physically consistent predictions. The hybrid approach leverages the robustness of numerical methods and the adaptability of machine learning, reducing computational cost and improving accuracy. The framework is validated on benchmark problems involving coupled transport phenomena, demonstrating improved convergence and predictive performance. This work highlights the potential of AI-augmented numerical techniques for next-generation fluid and thermal system simulations.	Dr. Sapna Sharma	8591686627	sapna.sharma@thapar.edu	Dr. Avinash Chandra, Dr. Sharad Saxena	avinash.chandra@thapar.edu, sharad.saxena@thapar.edu	9914255295	CHED

7	IoT-Driven Smart Traffic Management System for Urban Congestion Reduction	Objective: Design and implement a real-time traffic monitoring and management system using IoT sensors, AI algorithms, and cloud computing to reduce traffic congestion in a smart city. Scope: •Develop a prototype to capture real-time traffic data using sensors and cameras. •Process data using edge computing and AI for traffic prediction. •Implement an adaptive traffic light management algorithm. •Provide a user dashboard for monitoring and analysis. Tools and Technologies: •Hardware: Arduino/Raspberry Pi/Jetson Nano, IR Sensors, Cameras •Software: Python, Node-RED, MATLAB •AI Algorithms: Machine Learning (Regression, Classification), Computer Vision •Cloud Platforms: AWS, Azure, or Google Cloud •Visualization: Power BI or Tableau	Dr. Amit Kumar trivedi	07719669340	amitkumar.trivedi@thapar.edu	Dr. Rajesh Mehta	rajesh.mehta@thapar.edu	09911410228	CSED
8	Low-Power, Real-Time TinyML-Based ECG Monitoring System with Event-Driven Cloud Communication	Cardiovascular diseases often require continuous heart monitoring, but traditional systems are bulky and require manual analysis. This project proposes a lightweight, wearable ECG monitoring solution that performs 'real-time arrhythmia detection on-device' using TinyML, while minimizing power consumption through event-driven data transmission to Cloud. The final product is a "Band-Aid" style disposable patch using an ESP32 and a single-lead ECG sensor.	Dr. Anil Singh	9882488483	anil.singh@thapar.edu	Jaspinder Kaur	jaspinder.kaur@thapar.edu	8872561070	CSED
9	Design and Implementation of a Privacy-Preserving Federated Learning Framework for Healthcare Analytics	This ELC project focuses on the design and implementation of a privacy-preserving healthcare analytics system using Federated Learning (FL). The objective is to enable multiple clients (such as hospitals or edge devices) to collaboratively train machine learning models without sharing sensitive patient data. The project explores challenges such as data heterogeneity, client synchronization, and reduced model accuracy in traditional FL systems. To address these issues, the proposed work investigates advanced approaches like Asynchronous Peer-to-Peer Federated Learning (AP2PFL) and hybrid architectures combining deep learning (CNN-based feature extraction) with classical machine learning models. The system includes modules for local model training, feature extraction, and centralized aggregation of model weights. Experimental evaluation will compare different architectures to improve performance metrics such as accuracy and recall. This project aims to develop efficient, scalable, and secure solutions for real-world healthcare applications while ensuring data privacy and reliability.	Dr. Anjula Mehto	8839235288	anjula.mehto@thapar.edu	Dr. Saif Nalband	saif.nalband@thapar.edu	9322317570	CSED

10	AI-Driven Smart ICU Companion System for Patient Monitoring, Communication, and Predictive Healthcare Analytics	Modern intensive care units (ICUs) primarily focus on sustaining physiological stability, often overlooking patients' emotional well-being, communication challenges, and predictive health insights. This project proposes a multimodal AI-powered Smart ICU Companion System that integrates real-time physiological monitoring, emotional intelligence, and data-driven predictive analytics. The system combines computer vision, machine learning, and sensor-based data acquisition to enable non-verbal communication (e.g., blink/eye-gaze interfaces), emotion recognition, and continuous patient monitoring. A key component is a family-facing dashboard that provides simplified real-time updates, improving transparency and emotional connectivity. The system also includes a smart ICU environment module, which dynamically adjusts lighting, sound, and ambience based on patient emotional and physiological states. By combining human-centered care with data mining and predictive intelligence, this solution aims to transform ICU care into a more holistic, responsive, and intelligent ecosystem.	Dr. Anu Bajaj	7988019480	anu.bajaj@thapar.edu	Dr. Deep Mann	deep.mann@thapar.edu	9465964609	CSED
11	XAI-Driven Intelligent Lung Analytics: A Synergistic Fusion of Attention-Guided Segmentation and Wavelet-Enhanced Deep Learning for Precision Disease Profiling	Lung diseases such as COVID-19, pneumonia, and pulmonary fibrosis pose significant challenges in clinical diagnosis due to the variability in infection patterns and the complexity of medical imaging data. Accurate identification of infected regions and assessment of disease severity are crucial for timely treatment planning and patient management. However, manual analysis of chest X-ray (CXR) and computed tomography (CT) images is time-consuming, subjective, and dependent on expert radiologists. To address these challenges, this project proposes a deep learning-based framework for automated lung disease analysis that integrates segmentation and classification tasks in a unified pipeline. Initially, the input medical images undergo preprocessing, including normalization, resizing, and lung region extraction. A segmentation model based on an Attention U-Net architecture is employed to accurately delineate infected lung regions from CT/CXR images. To enhance feature representation, the proposed framework incorporates wavelet-based fusion (Haar Wavelet Transform) to capture both spatial and frequency-domain information. The fused features are then utilized by a CNN-based classification network to categorize the disease severity into different levels (e.g., mild, moderate, severe). Furthermore, to improve the interpretability and clinical trustworthiness of the model, an explainability module using Grad-CAM is integrated to highlight the regions contributing to the classification decisions. This enables visual validation by clinicians and supports transparent decision-making. The performance of the proposed framework will be evaluated using standard metrics such as Dice Similarity Coefficient (DSC) for segmentation and Accuracy, Precision, Recall, and F1-score for classification. The proposed system aims to provide a reliable, efficient, and explainable solution for automated lung disease diagnosis, facilitating improved clinical outcomes. Objective 1 To develop a robust and clinically adaptable preprocessing framework for CT and CXR images that standardizes heterogeneous data through advanced normalization, lung region extraction, and resolution alignment, ensuring reliable and consistent input for deep learning-based analysis.	Dr. Chinmay Panigrahy	09124492547	chinmaya.panigrahy@thapar.edu	Paluck Arora	paluck.arora@thapar.edu	9306292557	CSED

12	Ethical Hacking: Decentralized cognitive security Framework	This framework introduces a self-evolving security architecture that harmonizes advanced AI reasoning with decentralized data integrity. By deploying Large Language Models (LLMs) at the edge, the system performs deep semantic audits to detect sophisticated anomalies and malicious intent that bypass traditional filters. These insights are then anchored to a blockchain-backed ledger, creating a transparent, immutable record of entity reputations. This dual-layered approach replaces centralized, easily compromised databases with a global "immune system," ensuring that once a threat is identified, its signature is permanently recorded, providing a tamper-proof foundation for autonomous trust in digital ecosystems	Dr. Gurpal Singh Chhabra	08968889740	gurpal.singh@thapar.edu	Dr. Aashima Sharma	aashima.sharma@thapar.edu	9876728411	CSED
13	AI Based Approach for Female Health Assessment using Imaging and Physiological Signals	This proposal presents an AI-based framework for female health assessment using heterogeneous data sources. The system leverages imaging data along with complementary physiological signals to learn patterns associated with underlying health conditions. By integrating data-driven modeling techniques, the framework aims to capture complex relationships between observable inputs and latent health indicators. The approach emphasizes robustness, scalability, and adaptability to limited and variable data conditions. Machine learning models will be developed to enable early risk assessment and support informed decision-making. The study also focuses on evaluating model performance using standard metrics and analyzing generalization across different data scenarios. The outcome of this work is expected to contribute toward intelligent, non-invasive, and personalized health assessment systems, with potential applications in preventive healthcare and clinical decision support.	Dr. Jyoti Maggu	7988392850	jyoti.maggu@thapar.edu	Simranjit Kaur	simranjit.kaur@thapar.edu	7087030653	CSED

14	Illuminating the Heart: XAI-Driven Deep Learning for Transparent Cardiovascular Diagnosis and Risk Assessment	Cardiovascular diseases (CVDs), including coronary artery disease, arrhythmias, and heart failure, remain the leading cause of global mortality, often due to delayed diagnosis and complex clinical manifestations. Accurate and early detection is critical for improving patient outcomes; however, traditional diagnostic approaches relying on electrocardiograms (ECG), echocardiography, and cardiac imaging are often time-intensive, expertise-dependent, and prone to inter-observer variability. To address these limitations, this project proposes an Explainable Artificial Intelligence (XAI)-driven deep learning framework, termed DeepCardio-XAI, for automated cardiovascular disease detection and risk stratification. The framework integrates multi-modal data sources, including ECG signals and cardiac imaging (e.g., echocardiograms or MRI), enabling comprehensive cardiac assessment. Initially, raw input data undergo advanced preprocessing, including noise filtering (for ECG), normalization, segmentation of cardiac cycles, and region-of-interest extraction for imaging modalities. A hybrid deep learning architecture is then employed, combining CNNs for spatial feature extraction and temporal models (e.g., LSTM/Transformers) for sequential signal analysis. For imaging data, an Attention U-Net-based segmentation module is utilized to delineate critical cardiac structures such as ventricles and atria. To enhance representational richness, the framework incorporates multi-resolution feature fusion, potentially leveraging wavelet transforms or attention-based fusion mechanisms to capture both temporal dynamics and spatial hierarchies. The extracted features are then fed into a classification module to identify cardiovascular conditions and stratify disease severity or risk levels. A key component of the proposed system is its interpretability, achieved through XAI techniques such as Grad-CAM (for imaging) and SHAP/LIME (for tabular or signal data). These methods provide visual and quantitative explanations,	Dr. Paluck Arora	9306292557	chinmaya.panigrahy@thapar.edu	Chinmaya Panigrahy	chinmaya.panigrahy@thapar.edu	09124492547	CSED
15	Long-range Cross-Radio Communication using LoRa Modules	This project proposes a decentralized, off-grid communication system designed for military operations in areas lacking reliable infrastructure. It leverages the Reticulum networking stack to enable secure, encrypted communication across heterogeneous devices without relying on cellular networks or satellites. The system uses LoRa (Long Range Radio) for low-power, long-distance communication and Raspberry Pi-based bridge nodes to connect different network interfaces such as Wi-Fi and LoRa. Each device acts as both a user and a relay, forming a resilient multi-hop network that adapts dynamically to node failures or movement. Built-in cryptographic identities ensure secure, anonymous communication. This system enhances operational reliability by enabling real-time messaging, location sharing, and coordination in remote or hostile environments, making it highly valuable for border security, reconnaissance, disaster response, and mission-critical communication scenarios.	Dr. Prashant Singh Rana	9313889932	prashant.singh@thapar.edu	Dr. Anjula Mehto	anjula.mehto@thapar.edu	8839235288	CSED

16	Development of algorithm for detection and identification of poisonous gases in chemical clouds	This proposal aims to develop a portable, AI-based multi-sensor system for the detection and identification of poisonous gases in chemical clouds. Traditional gas detection methods rely on single sensors, which suffer from cross-sensitivity and limited accuracy in mixed-gas environments. To address this, the proposed solution utilizes a sensor array (MQ series) integrated with an ESP32 microcontroller to enable real-time data acquisition and wireless transmission. The core innovation lies in applying machine learning with Sensor Fusion, where data from multiple sensors is processed to identify specific gases and estimate toxicity levels. This enhances reliability under uncertain and noisy environmental conditions. The system is designed to be low-cost, energy-efficient, and portable, making it suitable for field deployment in hazardous environments. This approach provides a scalable and intelligent solution aligned with modern defense and disaster-response requirements, offering real-time monitoring, improved accuracy, and rapid decision-making capability.	Dr. Prashant Singh Rana	9313889932	prashant.singh@thapar.edu	Dr. Saif Nalband	saif.nalband@thapar.edu	9503516325	CSED
17	A Hybrid AI Approach for Smart Healthcare: Enhancing Disease Prediction and Clinical Decision Support	This work proposes a hybrid AI-based smart healthcare system that integrates multiple machine learning and deep learning techniques to enhance disease prediction and clinical decision support. The system combines data-driven models such as neural networks, ensemble learning, and probabilistic methods to analyze heterogeneous healthcare data, including patient history, diagnostic reports, and real-time sensor inputs. By leveraging hybrid intelligence, the framework improves prediction accuracy, robustness, and adaptability across diverse medical conditions. Additionally, the system incorporates a decision support module that assists clinicians by providing evidence-based recommendations and risk assessments. The proposed approach aims to reduce diagnostic errors, enable early disease detection, and support personalized treatment planning. Overall, the framework contributes to the development of intelligent, scalable, and efficient healthcare solutions aligned with modern digital health ecosystems.	Dr. Rajendra Kumar Roul	9834112291	raj.roul@thapar.edu	Dr. Ravneet Kaur	ravneet.kaur@thapar.edu	8146732490	CSED
18	Real-Time Vision-Based Defect Detection for Large-Scale Additive Manufacturing	Large-scale on-site additive manufacturing suffers from defects such as layer misalignment, over/under-extrusion, nozzle clogging. Currently, defect detection is mostly manual leading to material waste, structural weakness, increased cost. There is a need for a real-time automated vision system that detects defects during printing. So, the primary objective of the project will be to develop a computer vision-based system for real-time defect detection to detect multiple defect types.	Dr. Ravneet Kaur	8837721772	ravneet.kaur1@thapar.edu	Dr. Malkeet Singh	malkeet.singh@thapar.edu	8968928466	CSED
19	Automated Recognition and Digitization of Handwritten Circuit Schematics	This project aims to develop an intelligent system for the digitization of handwritten circuit diagrams into structured, machine-readable formats. Engineers and students often sketch circuits on paper, which are difficult to store, edit, or simulate. The proposed solution leverages computer vision and deep learning to automatically recognize circuit components (e.g., resistors, capacitors, transistors) and their interconnections from scanned images or photographs. The system will include modules for image preprocessing, symbol detection, wire extraction, and topology reconstruction. The project emphasizes accuracy, robustness to varying handwriting styles, and real-time usability. Expected outcomes include a working prototype, annotated dataset, and performance evaluation.	Dr. Ravneet Kaur	8837721772	ravneet.kaur1@thapar.edu	Dr. Mandeep Kaur	mandeep.bajwa@thapar.edu	8437533551	CSED

20	UAV Based Indoor Navigation	This proposal focuses on developing a UAV-based indoor navigation system capable of autonomous operation in GPS-denied environments. The system integrates onboard vision sensors, depth estimation, and lightweight SLAM (Simultaneous Localization and Mapping) algorithms to enable real-time mapping and obstacle avoidance. Advanced path planning and control strategies will be implemented to ensure stable and efficient navigation in complex indoor spaces such as warehouses, industrial facilities, and disaster-hit buildings. The UAV will be designed for low-cost embedded deployment, ensuring scalability and energy efficiency. The project also explores sensor fusion using vision, inertial measurement units (IMU), and ultrasonic sensors to enhance localization accuracy. The proposed solution aims to support applications including search and rescue, surveillance, inventory monitoring, and smart infrastructure inspection, contributing to the advancement of autonomous aerial robotics in constrained environments.	Dr. Sachin Kansal	9873153996	sachin.kansal@thapar.edu	Sachin Kansal	sachin.kansal@thapar.edu	9873153996	CSED
21	AI powered non-invasive and low-cost diagnostic system for knee-joint disorders	In India, over 150 million people suffer from knee problems, with four million requiring total knee replacement due to osteoarthritis (OA). However, current non-invasive diagnostic methods such as X-rays, CT scans, and MRIs are costly, leading to the need for an alternative diagnostic method. Vibroarthrographic (VAG) signals provide valuable information about the pathological and physiological state of the knee joint and can be acquired by placing a sensor at the mid-patella position of the knee and instructing the subject to perform flexion and extension movements. To address this issue, we propose the development of a real-time, low-cost, and non-invasive diagnostic system for OA using VAG signals. The system would acquire VAG signals and utilize machine learning for pattern classification to detect abnormal VAG signals. The proposed methodology would be integrated and deployed on hardware as a device for early diagnosis of OA, providing a low-cost and non-invasive alternative to current diagnostic methods. This work will result in a product that can assist physicians in making efficient and reliable clinical decisions at local clinics, thereby reducing healthcare expenses.	Dr. Saif Nalband	9322317570	anjula.mehto@thapar.edu	Dr. Anjula Mehto	anjula.mehto@thapar.edu	8839235288	CSED
22	Quantum-Driven 3D Medical Image Classification for Healthcare Intelligence	3D medical imaging is indispensable for contemporary diagnosis, treatment planning, and longitudinal disease monitoring, as it provides rich volumetric information about anatomical structures and pathological changes. Yet, deriving clinically meaningful insights from such high-dimensional data remains computationally intensive and methodologically challenging for conventional approaches. Quantum computing offers a transformative computational paradigm rooted in superposition and entanglement, enabling the representation, processing, and analysis of complex data structures beyond the limits of classical computation alone. Although large-scale practical quantum advantage in medical imaging is still at an early stage, the exploration of hybrid quantum-classical frameworks holds strong promise for more efficient feature encoding, improved representation learning, and enhanced classification of 3D medical images. This proposal, therefore, aims to initiate a focused institutional research effort to investigate how quantum computing concepts can be integrated into 3D medical image classification, with the broader goal of advancing future intelligent, robust, and clinically impactful healthcare technologies.	Dr. Satbir Singh	9464281274	satbir.singh@thapar.edu	Dr. Amanpreet Kaur	aman.preet@thapar.edu	9855014327	CSED

23	DriveSafeAI 2.0: A Multimodal Edge-AI Framework for Real-Time Driver Drowsiness Detection and Intelligent Transportation Safety	DriveSafeAI 2.0 is a multimodal edge-AI framework designed to detect driver drowsiness in real time by integrating visual, behavioral, and sensor-based data. It leverages on-device intelligence for low-latency processing, ensuring timely alerts and enhanced road safety. The system aims to contribute to intelligent transportation by reducing fatigue-related accidents through scalable and cost-effective deployment.	Dr. Shailendra Tiwari	9598404656	shailendra@thapar.edu	Manju Khurana	manju.khurana@thapar.edu	8824955448	CSED
24	Speech and Facial based emotion detection using VR and Machine Learning	Speech-based emotion detection using machine learning is an emerging area of research that focuses on identifying human emotions from vocal signals. It involves analyzing various acoustic features such as pitch, tone, energy, and speech rate to classify emotions like happiness, sadness, anger, and neutrality. Feature extraction techniques such as Mel-Frequency Cepstral Coefficients (MFCC) and Mel-spectrograms are commonly used to capture both spectral and temporal characteristics of speech signals. Machine learning and deep learning models, including Support Vector Machines (SVM), Random Forest, and Convolutional Neural Networks (CNN), are employed to learn patterns associated with different emotional states. Data augmentation and noise reduction techniques further enhance model robustness and accuracy. This technology has wide applications in human-computer interaction, mental health monitoring, customer service, and virtual assistants, making systems more intuitive and responsive to human emotions.	Dr. Sharad Saxena	8195098711	sharad.saxena@thapar.edu	Dr. Samya Muhuri	samya.muhuri@thapar.edu	9038720709	CSED
25	Artificial Intelligence-Based System for Breast Cancer Detection	This project focuses on developing a breast cancer detection system using annotated imaging datasets. The methodology includes image preprocessing, region-of-interest extraction, and feature representation, followed by implementation of machine learning models for classification. Comparative analysis of different algorithms will be performed to identify optimal approaches. The outcome will be a validated prototype capable of distinguishing benign and malignant cases with improved reliability and computational efficiency.	Dr. Simranjit Kaur	7087030653	simranjit.kaur@thapar.edu	Dr. Sachin Kansal	sachin.kansal@thapar.edu	9873153996	CSED
26	WiTrack: Tracking human activity using Wifi routers	Traditional human activity detection systems rely heavily on cameras, wearable devices, or dedicated sensors, which introduce challenges such as privacy concerns, user inconvenience, limited scalability, and higher deployment costs. In environments like homes, hospitals, and offices, continuous monitoring using these methods is often impractical or intrusive. Recent advances show that WiFi signals inherently capture environmental changes, as human movement alters signal propagation characteristics such as Channel State Information (CSI). However, leveraging WiFi signals for reliable activity recognition presents several challenges: Extracting meaningful features from noisy and high-dimensional CSI data Ensuring robustness across different environments and users Achieving real-time performance on resource-constrained edge devices Maintaining high accuracy without extensive labeled datasets Therefore, the problem is to design and implement a low-cost, privacy-preserving, device-free sensing system that uses WiFi CSI data to accurately detect and classify human activities in real time, while being robust, scalable, and deployable on commodity hardware such as ESP32 devices.	Dr. Subhrendu Chattopadhyay	9435658234	subhrendu.chattopadhyay@thapar.edu	Dr. Raman Kumar Goyal	ramankumar.goyal@thapar.edu	+91 94632 79788	CSED
27	Metasurface Absorber for Bunker Design and Stealth Applications	This project focuses on designing a metasurface absorber tailored for bunker applications, enabling broadband electromagnetic absorption, structural stealth, and resilience against high-power microwave threats. The design leverages parametric optimization to enhance key performance metrics including absorptivity, bandwidth, angular stability, and structural integration while ensuring mechanical robustness for defense-grade deployment.	Dr Rajesh Khanna	9872883263	rkhanna@thapar.edu	Dr. Aarti Bansal	aarti.bansal@thapar.edu	9855856504	ECED

28	AI-Enabled Design and Optimization of Silver/Graphene Ink-Based Printed MIMO Antenna for 5G Communication	The rapid advancement of 5G systems has created a strong demand for compact, flexible, and high-performance antenna technologies capable of supporting high data rates and reliable multi-channel communication. In this context, Silver/Graphene ink-based printed antennas have emerged as a promising solution, offering a combination of high electrical conductivity from silver and mechanical flexibility and chemical stability from graphene. These properties make them particularly suitable for next-generation flexible and wearable wireless devices	Dr Rajesh Khanna	9872883263	rkhanna@thapar.edu	Dr. Navneet Kaur	navneet_kaur@thapar.edu	8146659461	ECED
29	Design and Development of a Multi-Functional IoT-Based Signal Analysis and Generation System	This project presents the design and development of a compact, low-cost, and multi-functional IoT-based system capable of both signal generation and signal analysis. The system integrates a microcontroller-based architecture with SPI-interfaced analog peripherals to implement a function generator (FG) and a digital storage oscilloscope (DSO) within a single platform. By employing bare-metal programming techniques, deterministic timing control is achieved, enabling efficient waveform synthesis and real-time data acquisition. The analog front-end is designed to handle a wide input range, ensuring reliable signal conditioning and measurement. IoT connectivity further enhances the system by allowing remote monitoring, control, and data visualization. The key novelty of this work lies in integrating both a DSO and a Function Generator into a single embedded system, which is typically not achievable with conventional microcontroller-based approaches without dedicated instrumentation hardware. This makes the system a cost-effective and educational alternative to standalone laboratory instruments.	Dr. Amanpreet Kaur	9855014327	aman.preet@thapar.edu	Dr. Anil Singh and Dr Manu Bansal	anils.rawat@thapar.edu , mbansal@thapar.edu	98885 86454	ECED

30	AI-Driven Self-Tuning Implantable Antenna for Adaptive Biomedical Monitoring	Implantable biomedical devices require reliable wireless communication despite dynamic variations in the human body environment. This work presents an AI-driven self-tuning implantable antenna integrated with anomaly detection and parameter monitoring for adaptive biomedical applications. The proposed system employs a compact microstrip implantable antenna whose performance is influenced by tissue heterogeneity and physiological changes. A machine learning model is developed to continuously monitor key antenna parameters such as resonant frequency, return loss, and impedance, enabling the detection of anomalies including detuning, faults, and performance degradation. Upon detection, the system predicts optimal tuning parameters to restore antenna performance in real time. A dataset is generated under varying dielectric conditions to train and validate the model. Experimental results demonstrate that the system can accurately identify faults and parameter variations while maintaining improved impedance matching and stable operation compared to conventional designs. The system can further support IoT-based alert mechanisms for intelligent and remote health monitoring. 🔹 What's Now Strong in Your Abstract ✓ Clearly mentions fault detection as a RESULT ✓ Includes parameter change monitoring ✓ Shows system capability (identify + correct + alert) ✓ Fully aligned with research publication expectations If you want next: ✓ Keywords (important for indexing) ✓ Title + objectives + novelty (for proposal/paper) ✓ Block diagram (very important for viva/presentation) Just tell me 🐼	Dr. Amanpreet Kaur	9855014327	aman.preet@thapar.edu	Dr. Jaswinder Kaur	jaswinder.kaur@thapar.edu	99888 00901	ECED
31	Machine Learning-Assisted Optimization of Quantum Communication Channels	This proposal aims to explore machine learning-assisted optimization techniques for enhancing the performance of quantum communication channels. The study will investigate how supervised and reinforcement learning models can be integrated with quantum systems to dynamically optimize key parameters such as channel noise, error rates, and resource allocation. By simulating quantum channels under realistic conditions, the model will learn optimal strategies to maximize fidelity, channel capacity, and secure key rates in protocols like quantum key distribution and entanglement-based communication. The framework will incorporate hybrid classical-quantum approaches, leveraging tools such as Qiskit for quantum circuit simulation and standard machine learning libraries for training and evaluation. The expected outcome is an intelligent optimization framework that improves reliability, efficiency, and scalability of quantum communication systems, contributing to the advancement of secure and adaptive next-generation quantum networks.	Dr. Amit Mishra	9977642640	amit_mishra@thapar.edu	Dr. Ashutosh Kumar singh	aksingh@thapar.edu	9753114302	ECED

32	Development and simulation of Quantum Communication Networks Using Qiskit	This proposal focuses on the development, simulation and analysis of quantum communication networks using Qiskit. The objective is to model key quantum communication protocols such as quantum key distribution (QKD), quantum teleportation, and entanglement swapping within a scalable network framework. The study will incorporate realistic channel conditions, including noise, decoherence, and transmission losses, to evaluate system performance. Using Qiskit's circuit-based simulation capabilities, the work will analyze critical metrics such as fidelity, error rates, and key generation efficiency. The proposed approach will also explore the role of quantum repeaters and error mitigation techniques in improving long-distance communication. The outcomes are expected to provide insights into the design and optimization of robust and secure quantum communication networks, contributing to the development of next-generation quantum internet infrastructure.	Dr. Amit Mishra	9977642640	amit_mishra@thapar.edu	Dr. Ashutosh Kumar Singh	aksingh@thapar.edu	9753114302	ECED
33	Intelligent Signal Conditioning Circuit for Real-Time IoT Air Quality Monitoring	This proposal focuses on the design and development of a high-performance signal conditioning circuit for accurate air quality monitoring integrated with IoT. Gas sensors typically produce weak, nonlinear, and noise-prone signals, making signal conditioning a critical component. The proposed system emphasizes amplification, precise filtering, impedance matching, and linearization using optimized operational amplifier configurations. The conditioned signals are digitized through a high-resolution ADC and processed by a microcontroller (e.g., ESP32) for reliable data acquisition. IoT connectivity enables real-time transmission of calibrated air quality data to cloud platforms. A data logging unit ensures continuous storage for long-term analysis, while a graphical interface displays real-time and historical trends. By prioritizing robust signal conditioning, it significantly enhances measurement accuracy, stability, and sensitivity. The system aims to provide a cost-effective, scalable solution for smart environmental monitoring, supporting applications in smart cities, industrial safety, and pollution control.	Dr. Anil Arora	8437701902	anil.arora@thapar.edu	Dr. Surbhi Sharma	surbhi.sharma@thapar.edu	9815302660	ECED
34	Development of High Speed Data Communication Circuits	In this, we want to design and develop circuits for high-speed data communication using off-the-shelf components and preferably FPGA in order to develop customised and cost effective solutions. This can be used in UAV video transceivers etc.	Dr. Anil Singh	9888586454	anils.rawat@thapar.edu	Dr. Manu Bansal and Dr. Aman preet-II	mbansal@thapar.edu and aman.preet@thapar.edu	9417162059 and 9855014327	ECED
35	SDR based JSAC system	Joint sensing and communication (JSAC) turns out to be a prospective area in 6G communication especially for autonomous vehicle. In this a single system will serve the demand of sensing and communication both. Particularly this project aims to analysis of a single waveform for JSAC system. However this design will require to meet certain tradeoffs as the demand of sensing and communication contradicts each other.	Dr. Ashutosh Kumar Singh	9753114302	aksingh@thapar.edu	Dr. Hem Dutt Joshi	hemdutt.joshi@thapar.edu	8727871864	ECED

36	Deep Learning Framework for Early-Stage Ischemic Stroke Identification Using Multimodal Clinical Data	CNNs to extract and analyze subtle early ischemic changes from CT scans — such as hyperdense artery signs, loss of gray-white matter differentiation, and early infarct patterns — which are often missed during manual clinical review. Simultaneously, structured lifestyle risk factors including hypertension history, smoking habits, diabetes status, physical activity levels, BMI, and dietary patterns are encoded and processed through fully connected neural network layers. A multimodal fusion architecture combines imaging features with lifestyle-derived risk indicators to produce a unified predictive model capable of classifying stroke risk at an early stage with improved sensitivity and specificity. The model is trained and validated on curated clinical datasets, with performance evaluated using metrics such as AUC-ROC, accuracy, sensitivity, and specificity. The sensor inputs for real time time series data shall be taken from wearable sensor based device.	Dr. Gaganpreet Kaur	8146560369	gaganpreet.kaur@thapar.edu	Dr. Sandeep Mandia	sandeep.mandia@thapar.edu	9799790044	ECED
37	AR enabled Underwater Scuba Diver Safety Device	A wrist-worn/ Head display device that monitors depth, pressure, and uses ultrasonic sensing to detect underwater obstacles, alerting the diver via vibration and LED warnings	Dr. Gaganpreet Kaur	8146560369	gaganpreet.kaur@thapar.edu	Dr. Anil Arora	anil.arora@thapar.edu	8437701902	ECED
38	Artificial Intelligence Approaches to Image Understanding	The role of Artificial Intelligence in image understanding has evolved significantly with advances in Computer Vision and deep learning. This topic focuses on developing intelligent systems capable of interpreting complex visual data across domains such as healthcare, surveillance, and autonomous systems. The scope includes image classification, object detection, segmentation, and multimodal learning by integrating images with text or sensor data. The importance lies in enabling accurate, real-time decision-making and reducing human intervention, particularly in critical applications like medical diagnosis and smart infrastructure. Incorporating explainability ensures transparency and trust in AI systems.	Dr. Hardeep singh	9855616892	neeru.jindal@thapar.edu	Dr. Kulbir Singh, Dr Arun Pundhir, Dr Neeru Jindal	Ksingh@thapar.edu, arun.pundhir@thapar.edu, neeru.jindal@thapar.edu	9465060700	ECED

39	Development of Smart Autonomous Mobile Robot for Industrial Applications	Imagine building a robot that navigates a real indoor environment on its own — mapping its surroundings, avoiding obstacles, and delivering materials precisely where they're needed. This summer, you'll do exactly that. You'll work as part of a hands-on engineering team to design, build, and field-validate an industrial-grade Autonomous Mobile Robot (AMR) equipped with integrated storage drawers for payload delivery. The robot will make intelligent navigation decisions by fusing data from multiple sensors — LiDAR, cameras, and additional onboard sensors — processed through a Raspberry Pi as the central computing brain and coordinated with low-level motor control via an Arduino Uno. What you'll gain hands-on experience with: ROS (Robot Operating System) — the industry-standard framework for robot software Multi-sensor fusion — combining LiDAR, camera, and other sensor streams for reliable navigation; Embedded systems — programming and integrating Raspberry Pi and Arduino Uno in a real robot; Autonomous navigation — mapping, localization, and real-time path planning in dynamic environments The robot will be tested and validated in a real indoor environment, giving you the rare experience of seeing your engineering work perform beyond the lab. This project is ideal for students interested in robotics, embedded systems, computer vision, or autonomous vehicles — and who want practical, industry-relevant experience that goes well beyond the classroom.	Dr. Hem Dutt Joshi	8727871864	hemdutt.joshi@thapar.edu	Dr. Kulbir Singh	ksingh@thapar.edu	9417410974	ECED
40	Smart Antennas for Smart Networks: AI-Driven Multi-Band MIMO Design for real-world 5G/6G Use Cases	This proposal focuses on the design and development of intelligent multi-band MIMO antenna systems using AI techniques for real-world 5G/6G applications. The project aims to integrate machine learning with antenna design to optimize parameters such as bandwidth, gain, isolation, and efficiency across multiple frequency bands. Students will gain hands-on experience in simulation, prototyping, and testing using advanced tools and 5G lab infrastructure. The work will address practical use cases in healthcare, IoT, smart cities, and next-generation wireless communication systems. The project encourages innovation, interdisciplinary learning, and industry-relevant skill development, enabling students to contribute to cutting-edge wireless technologies and prepare for careers in future communication networks and research.	Dr. Jaswinder Kaur	9988800901	jaswinder.kaur@thapar.edu	Dr. Navneet Kaur	navneet_kaur@thapar.edu	8146659461	ECED
41	Powering Implants Wirelessly: Efficient RF Energy Harvesting Using Rectenna for Smart IoT Medical Implants	This proposal focuses on developing an efficient RF energy harvesting system using rectenna technology to wirelessly power low-power IoT-based medical implants. The project aims to design compact, biocompatible antennas integrated with high-efficiency rectifier circuits capable of converting ambient RF signals into usable DC power. Students will explore real-world healthcare applications such as wearable and implantable sensors for continuous health monitoring, reducing dependence on batteries and invasive replacements. The work includes simulation, design optimization, prototyping, and testing using available lab infrastructure. Emphasis will be placed on improving power conversion efficiency, miniaturization, and safe operation within biological environments. This project offers hands-on experience in cutting-edge wireless power transfer and biomedical applications, preparing students for research and innovation in next-generation healthcare technologies.	Dr. Jaswinder Kaur	9988800901	jaswinder.kaur@thapar.edu	Dr. Amanpreet Kaur	aman.preet@thapar.edu	9855014327	ECED

42	Smart Wastewater Surveillance and Analysis Framework	The Smart Wastewater Surveillance and Analysis Framework is an advanced system that monitors and evaluates wastewater quality in real time using modern sensing and communication technologies. The framework integrates multiple sensors to measure key parameters, including pH, turbidity, temperature, and dissolved oxygen, ensuring continuous data acquisition. Collected data is transmitted through IoT-enabled networks to a centralized platform for processing and analysis. The system employs data analytics techniques to detect anomalies, predict contamination levels, and generate timely alerts for preventive action. This approach enhances wastewater management efficiency, reduces environmental risks, and supports regulatory compliance. Additionally, the framework provides a user-friendly interface for visualization and decision-making. By enabling proactive monitoring and intelligent analysis, the proposed system contributes to sustainable water resource management and improved public health outcomes, making it suitable for urban, industrial, and rural ponds.	Dr. Karmjit Singh Sandha	9646020543	kssandha@thapar.edu	Dr. Shireesh Kumar Rai	skumar.ra@thapar.edu	9911941540	ECED
43	Digital bills, purpose, policy and practice using AI Ethics	Digital bills are electronic invoices and receipts managed through intelligent systems, increasingly supported by AI. Their purpose is to enhance efficiency, reduce paper usage, and enable automated financial tracking while aligning with ethical AI practices such as transparency, fairness, and accountability. The importance of digital bills in the context of AI ethics lies in ensuring secure, unbiased, and privacy-preserving financial transactions. AI-driven billing systems can detect anomalies, prevent fraud, and personalize services, but they must avoid data misuse and algorithmic bias. Ethical implementation builds user trust, ensures regulatory compliance, and promotes responsible data governance. The methodology involves deploying AI models for data extraction, validation, and fraud detection, integrated with secure cloud-based platforms. Privacy-preserving techniques like encryption and anonymization are essential. Additionally, organizations must follow ethical guidelines, conduct regular audits, and ensure explainability in AI decisions to maintain fairness, transparency, and accountability in digital billing systems.	Dr. Kulbir Singh	9417410974	neeru.jindal@thapar.edu	Dr. Hemdutt Joshi	hemdutt.joshi@thapar.edu	8727871864	ECED
44	Towards Battery-less circuits using RF energy harvesting	A battery-less circuit using RF (Radio Frequency) energy harvesting captures ambient electromagnetic signals and converts them into usable electrical power, enabling ultra-low-power devices to operate without batteries. The system typically consists of an antenna that collects RF signals from sources like Wi-Fi routers, mobile towers, or dedicated transmitters. These signals are alternating (AC) in nature and very low in power. The rectifier—often built using Schottky diodes—converts the RF AC signal into DC voltage. This DC output is then stored in capacitors or supercapacitors. A voltage regulator may be used to provide a stable output suitable for electronic components. Because harvested RF power is extremely small (μW to mW range), such circuits are designed for ultra-low-power applications, including IoT sensors, RFID tags, wearable devices, and biomedical implants.	Dr. Manu Bansal	9417162059	mbansal@thapar.edu	Anil Singh, Gitanjali Chandwani	anils.rawat@thapar.edu , gc.manocha@thapar.edu	9888586454, 9557406972	ECED

45	Artificial Intelligence for Early Prediction and Risk Stratification of Chronic Heart Failure	The application of Artificial Intelligence in predicting Chronic Heart Failure offers significant potential for early diagnosis and improved patient outcomes. This topic focuses on leveraging machine learning and deep learning models to analyze clinical, imaging, and wearable sensor data for accurate risk prediction and disease progression monitoring. The scope includes integrating electronic health records, ECG signals, and imaging data to develop robust predictive systems. The importance lies in reducing hospital readmissions, enabling timely interventions, and supporting personalized treatment strategies. Incorporating explainability ensures that predictions are transparent and clinically interpretable, increasing trust among healthcare professionals. A hardware and software integrated wearable device will be developed.	Dr. Neeru Jindal	9465060700	neeru.jindal@thapar.edu	Dr Hardeep	hardeep@thapar.edu	9855616892	ECED
46	RAG-based System for Information Retrieval from TIET Policy Documents	Often, a simple query from a university member requires significant effort to find the right document and obtain the right information. In the AI era, this can be automated using RAG systems, where documents are stored in databases, and AI algorithms can search and retrieve requested information. Documents are first divided into chunks, and their embeddings computed using LLM are stored in a vector database. Once all the documents are stored in the database, users can run queries. A query embedding is first computed using the LLM, and then the query embedding is compared with the stored documents in the vector database. Top-k similar embeddings are retrieved from the database. A prompt is structured in which the query is embedded alongside retrieved embeddings from the vector database as context. This prompt is passed to LLM for structured output of the query based on the given context. The LLM output is then given to the user for the query.	Dr. Sandeep Mandia	9799790044	sandeep.mandia@thapar.edu	Dr. Neeru Jindal	neeru.jindal@thapar.edu	9465060700	ECED
47	AI-Driven Mind-Wandering Detection in Learning Settings	MIND-WANDERING (MW) is the experience of thoughts not remaining on a single topic, particularly when people are engaged in an attention-demanding task. MW degrades the efficiency of learning if it appears when people are studying, especially during the COVID-19 pandemic, where online learning became a necessity. An AI-driven mind-wandering detection system shall be developed in this activity. Students shall learn about the entire life cycle of an AI project, from data collection, annotation, and model training to model deployment and hosting.	Dr. Sandeep Mandia	9799790044	sandeep.mandia@thapar.edu	Dr. Anil Arora; Dr. Kulbir Singh	anil.arora@thapar.edu; ksingh@thapar.edu	8437701902; 9417410974	ECED

48	Hardware Realization of High-Dimensional Chaotic Circuits for Secure Communication Systems	High-dimensional chaotic circuits provide an effective approach for secure communication due to their complex, unpredictable dynamics and large key space. Unlike low-dimensional systems, they offer enhanced security against signal reconstruction and interception. This work presents the hardware realization of such circuits using analog CMOS and discrete components. Nonlinear behavior is achieved through integrators, transconductance elements, and analog multipliers to generate hyperchaotic signals. These signals are used to mask information at the transmitter, ensuring secure data transmission. At the receiver, synchronization between matched chaotic circuits enables accurate recovery of the original signal. Achieving robust synchronization under practical conditions is a critical design consideration. The proposed system is optimized for low power, high-speed operation, and tolerance to noise and parameter variations. Experimental validation through phase portraits and Lyapunov exponents confirms chaotic behavior, making the system suitable for real-time secure communication in IoT and defense applications.	Dr. Shireesh Kumar Rai	9911941540	skumar.rai@thapar.edu	Dr. Karmjit Singh Sandha	kssandha@thapar.edu	9646020543	ECED
49	FPGA implementation of chaotic cryptographic algorithms for secure communication	In the rapidly evolving field of information security, cryptographic algorithms are essential for protecting sensitive data from unauthorized access and tampering. A key requirement for any cryptographic system is the generation of high-quality randomness. In this project, chaotic circuits will be designed, and their outputs will be utilized as a source of randomness for cryptographic applications. Subsequently, a secure communication model will be developed and implemented on an FPGA platform.	Dr. Sujit Kumar Patel	7879741990	sujit.patel@thapar.edu	Dr. Shireesh Kumar Rai	skumar.rai@thapar.edu	99119 41540	ECED

50	Live GPS tracking of drones using satellite data	Global Navigation Satellite Systems (GNSS), such as GPS, enable real-time positioning and navigation for a wide range of civilian applications, including unmanned aerial vehicles (UAVs or drones). The most widely used civilian signal is the L1 band (1575.42 MHz), which provides positioning information through satellite broadcasts. Live GPS tracking for drones using civilian L1 band signals relies on the fundamental principle of satellite-based positioning, where signals transmitted by navigation satellites are received by a GNSS module onboard the drone. The L1 band, operating at 1575.42 MHz, provides timing and positional data that the receiver processes to determine the drone's real-time location through trilateration. Since these signals are broadcast in a one-way manner, the drone itself computes its position rather than being actively tracked by satellites. To enable live tracking, this computed position must be transmitted to a ground station or remote system using an additional communication link such as radio telemetry or cellular networks. While it is technically possible to capture and process raw L1 signals using software-defined radio, practical implementations typically use dedicated GNSS modules for efficiency and reliability. Overall, the system combines satellite signal reception with onboard computation and external data transmission to achieve continuous real-time tracking of drones in civilian applications	Dr. Surbhi Sharma	9815302660	surbhi.sharma@thapar.edu	Dr.Kulbir singh	ksingh@thapar.edu	9417410974	ECED
51	Design of a Microwave Antenna for Targeted Energy Delivery	This project focuses on the design and development of a compact, efficient microwave antenna capable of delivering targeted energy within the ISM band. Emphasis will be placed on optimizing antenna performance parameters such as impedance matching, gain, and radiation behaviour, while also analyzing how electromagnetic waves interact with different materials	Rajesh Khanna	9872883263	rkhanha@thapar.edu	Dr Nitika Sharma	nitika.sharma@thapar.edu	8427928874	ECED
52	Microwave Wireless Power Transfer at 2.45 GHz using Rectenna Array for RF Energy harvesting for Low Power Devices	This project aims to develop an efficient microwave wireless power transfer (WPT) system operating at 2.45 GHz for powering low-power IoT devices. A compact rectenna (antenna + rectifier) will be designed and extended into an array configuration to enhance RF-to-DC conversion efficiency.	Rajesh Khanna	9872883263	rkhanha@thapar.edu	Dr Geetanjali	geetanjali@thapar.edu	7986505412	ECED
53	Private Cloud for secure drones flight	This project is on development of private cloud infrastructure for secure personal drone flying in short range and for local communication.	Dr. Kulbir Singh	9417410974	ksingh@thapar.edu	Dr. Hem Dutt Joshi	hemdutt.joshi@thapar.edu	8727871864	ELC
54	IoT enabled high frequency data mapping using force sensors	This Project is on development of a IoT enabled high frequency data mapping using force sensor.	Dr. Kulbir Singh	9417410974	ksingh@thapar.edu	Dr Ravinder Kumar Duvedi, Dr. Anil Arora,	anil.arora@thapar.edu, rduvedi@thapar.edu	8437701902	ELC
55	Robotic Monkey	We plan to build a AI integrated bio-inspired monkey robot that replicates primate agility and adaptability, enabling autonomous navigation through complex urban and natural environments where conventional robots fail.	Dr. Manu Bansal	9417162059	anils.rawat@thapar.edu	Dr. Anil Singh and Dr. Ravinder Kumar Duvedi	anils.rawat@thapar.edu and rduvedi@thapar.edu	9888586454 and 9780474711	ELC

56	Design and Fabrication of an Electromagnetic Levitation and Propulsion Vehicle	This project is on design and fabrication of a small prototype for proof of concept of an electromagnetic levitation and propulsion system.	Dr. Ravinder Kumar Duvedi	9780474711	rduvedi@thapar.edu	Dr Bikramjit Sharma, Dr Saharad Saxena, Dr Shakti Singh, Dr. Hem Dutt Joshi	bikramjit@thapar.edu	9417871105	ELC
57	A Unified, Transferable Robotic Control Board for Heterogeneous Robotic Platforms - Team G-Force	The purpose of this project (Universal Board for multiple robot) is to eliminate the redundant cycle of designing new electronics for every robot by creating a universal, recoverable control system. It aims to reduce electronic waste and development time by decoupling high-value processing hardware from the mechanical frame. Instead of building from scratch, developers use a single "brain board" that can be swapped between multiple robots—such as rovers, drones, or hexapods—allowing them to skip repetitive low-level wiring and focus on AI-driven mission objectives. The system works through a modular docking mechanism where the main control board, featuring a Teensy 4.1 and Raspberry Pi 5, slides into a robot's specific interface using pogo-pin connectors. Upon docking, the system reads a unique Hardware ID from the robot. This triggers the Raspberry Pi to automatically upload the correct motor logic and configuration settings to the Teensy. This "plug-and-play" architecture ensures that the electronics are never permanently fixed to a single machine, allowing for rapid redeployment across diverse robotic platforms.	Dr. Sharad Saxena	8195098711	sharad.saxena@thapar.edu	Dr. Ravinder Kumar Duvedi	jyoti.maggu@thapar.edu	7988392850	ELC
58	Physicobird: Game development using Unity Engine	This Project is on development of a game in the Unity engine.	Dr. Sharad Saxena	8195098711	sharad.saxena@thapar.edu	Dr. Sanjeev Bedi			ELC
59	Development of a Hybrid Stator Motor Integrated with Triboelectric Nanogenerator (TENG)-Based Wind Energy Harvesting System	This proposal outlines the design and development of a hybrid energy system that combines a stator-motor-based triboelectric nanogenerator module with a wind energy-harvesting module. The goal is to make energy use more efficient by using wind energy from the environment. This reduces the need for external power. The TENG module converts mechanical energy from wind-induced vibrations or rotations into electrical energy via contact electrification and electrostatic induction. The energy can be stored or used right away to power electronic devices. The proposed mechanism emphasises sustainable materials, compact integration, and enhanced reliability for small-scale applications, including automotive systems and remote devices. The system encourages the use of renewable energy and helps to create self-powered and energy-efficient electromechanical systems by combining conventional motor technology with advanced energy harvesting	Jayant Kolte	9960338240	rankit_phd21@thapar.edu	Dr. Ravinder Kumar Duvedi	rduvedi@thapar.edu	9780474711	ELC
60	Development of a Lab-scaled Oxy-hydrogen (HHO) gas generator using pencil cells	This proposal details a plan to build a small-scale oxy-hydrogen (HHO) generator using affordable pencil cells as electrodes. The main goal is to help students understand electrolysis better, using a potassium hydroxide (KOH) solution to make the water more conductive. Students will set up graphite electrodes from pencils inside a sealed chamber, then use direct current to break water down into hydrogen and oxygen gases. They'll collect the resulting HHO gas and test how well it burns and how much they've made. This project isn't just safe and budget-friendly—it gives students real, hands-on experience with green energy, electrochemistry, and alternative fuels. It also helps them see how sustainable technology works in practice, making the move away from fossil fuels feel much more real and achievable.	Dr. Amandeep Singh Oberoi	9988349747	oberoi@thapar.edu	Dr. Ravinder Kumar Duvedi	rduvedi@thapar.edu	9780474711	MED

61	VR Based Surgical Simulation for Gesture Rehearsal	This project aims to develop a simple prototype of an intra-operative surgical simulation system. The objective is to create a basic virtual environment where a user can visualize a 3D organ model and interact with it before performing a surgical action. Pre-recorded medical images (CT/MRI) will be used to generate a 3D model, which will be displayed using open-source software such as 3D Slicer or Unity. A basic interaction mechanism (VR/mouse or haptic device) will allow users to simulate actions like pushing or cutting. The system will demonstrate how real-time simulation can assist in planning surgical gestures safely. The project includes model generation, visualization, and simple interaction. This activity will help students understand the importance of simulation, image guidance, and safe surgical planning in modern healthcare.	Vivek Jain	9416156809	vivek.jain@thapar.edu	Dr. Tarunpreet Bhatia	tarunpreet@thapar.edu	8053302014	MED
62	Ground resolution enhancement for Low-Earth Orbit satellite data using image processing and machine learning techniques.	High-resolution imagery is a crucial part of providing an efficient solution to the issues weather monitoring, predictions, etc. However, using multiple high-resolution satellites will not be cost-effective and will also add to the space traffic. We propose to use the low-resolution image data to achieve higher ground resolution using image stacking and convolution techniques. Further, we plan to apply ML and neural network models to enhance the images, i.e., better ground resolution. The objective of the proposal is to design low-cost satellites for detailed real-time mapping of the aforementioned potent issues.	Dr. Mamta Gulati	08284903760	mamta.gulati@thapar.edu	Dr. Sumit Kumar	kumar.sumit@thapar.edu	9997500451	Others
63	Automation of satellite orbit manoeuvre countering the perturbation in Low Earth Orbits	Space weather variation and space debris are increasingly becoming a threat to the smooth sailing of satellites. Manual satellite manoeuvre from ground-based control centres is a tedious and sometimes not a fruitful way of doing the process. Automation of such processes is the need of the hour. Towards this end, we would like to propose a detailed modelling of orbit perturbation for LEO satellites due to SRP using Aditya L1 data. Next, we shall develop a semi-analytical algorithm for orbit manoeuvre calculation to counter satellite disintegration. The developed algorithm will be tested for compatibility with satellites' on-board processors.	Dr. Mamta Gulati	08284903760	mamta.gulati@thapar.edu	Dr. Vineet Srivastava	vineet.srivastava@thapar.edu	6239871878	Others
64	Mechanical Behavior Prediction of Additively manufactured IN625-SS316-IN718 Functionally Graded Materials	This project focuses on the development of a functionally graded material (FGM) using Inconel 625, Stainless Steel 316, and Inconel 718 through advanced additive manufacturing techniques. The aim is to achieve a gradual transition in composition to combine the superior high-temperature strength of Inconel 718, excellent corrosion resistance of Inconel 625, and cost-effectiveness of SS316. The study involves fabrication using Directed Energy Deposition, followed by evaluation of microstructure and mechanical properties. Optimization of process parameters will be carried out to enhance performance. The developed FGM is expected to exhibit improved resistance to thermal stress, oxidation, and mechanical properties making it suitable for aerospace and power generation sectors where multi-functional performance is required under extreme operating conditions.	Dr. Vineet Srivastava	9888031148	vineet.srivastava@thapar.edu	Dr. Malkeet Singh	malkee.singh@thapar.edu	8968928466	Others
65	Design of Inverse Piezoelectric Phenomenon based Nano-actuator	This Project is on development of nano-actuators for nano-drones from thin film polymeric materials using special 3D printing machine.	Dr. Ravinder Kumar Duvedi	9780474711	rduvedi@thapar.edu	Dr. Sudhir Kumar, Dr. Saharad Saxena	sudhir.kumar@thapar.edu	7888457989	ELC

66	Sign Link – Breaking Silence with Technology	Sign Link is an innovative application designed to reduce the communication gap between deaf and hearing people. Many deaf individuals struggle in daily life because sign language is not widely understood and real-time communication tools are limited. This project provides a simple and smart solution using Artificial Intelligence. □ The app offers interactive sign language learning, making it easy and interesting for users. It also includes a universal translator that converts speech, text, and sign language into understandable form. A special feature is real-time translation during video calls, which allows smooth two-way communication without needing an interpreter.	Dr. Sharad Saxena	8195098711	sharad.saxena@thapar.edu	NA	NA	NA	CSED
67	CAN Protocol for battery thermal management for THAR - EV Jeep	This Project is an on going ELC project where student team will develop and implement a CAN system for Thar EV-Jeep Project for status monitoring of various EV elements deployed in development of the Thar EV.	Dr Nirbhaw Jab Singh	9779379414	nirbhaw@thapar.edu	Dr Shakti Singh			EIED
68	CFD analysis of inverse piezo-electric actuator for nano-drones	Under this project students will be engaged in exploring the use of professional CFD tools to test the mechanical behaviour and CFD analysis of non-actuator designs to be used in nano-drones.	Dr Ravinder Kumar Duvedi	9780474711	rduvedi@thapar.edu	NA	NA	NA	ELC
69	Design of 3D CAD models and production drawings for a scaled prototype of a five-axis CNC machining center	Under this project we will be targeting to design and develop a prototype of a table-top 5-axis CNC machine.	Dr Ravinder Kumar Duvedi	9780474711	rduvedi@thapar.edu	NA	NA	NA	ELC
70	Design of Mechanical Structure and ECU integration for EV-Test Rig	This is an ongoing ELC project where this season students will work on the assembly, and testing of various available elements of EV test rig for testing the performance of Lithium Battery Pack and Electric Drive Motor.	Mr Sumit Sharma	9915599066	ssharma@thapar.edu	Dr Ravinder Kumar Duvedi, Dr Bikramjit Sharma, Dr Krishan Kumar Gupta, Dr Harishankar Singh	rduvedi@thapar.edu, bikramjit@thapar.edu	9417871105	ELC
71	Design and fabrication of a Stewart Platform	This project is to design and fabricate an inhouse model of a Stewart platform for introducing the students to parallel link robot, control and machine vision integration in robotics.	Dr Ravinder Kumar Duvedi	9780474711	rduvedi@thapar.edu	Dr Sanjeev Bedi			ELC
72	Design and Fabrication of a digital DIY Machine Kit for Tensile Testing of wire samples	This learning project focuses on the design and fabrication of a portable test machine integrated with a load cell and Arduino-based data acquisition system to evaluate the elastic modulus of metallic wire specimens. Students will engage in the complete engineering cycle, including conceptual design, mechanical fabrication, sensor integration, calibration, programming, and experimental validation. The system will measure load-deflection behavior during tensile tests and compute elastic modulus through real-time data processing. The project promotes hands-on learning in mechanics of materials, instrumentation, embedded systems, and digital manufacturing.	Dr. Deepak Jain	7837841951	deepak.jain@thapar.edu	Dr Ravinder Kumar Duvedi, Dr Bikramjit Sharma	bikramjit@thapar.edu	9417871105	ELC

73	Design and Fabrication of a digital DIY Machine Kit for Tensile Testing of 3D printed polymeric Dog-bone samples	This learning project focuses on the design and fabrication of a portable bending test machine integrated with a load cell and Arduino-based data acquisition system to evaluate the elastic modulus of 3D printing dog-bone shaped specimens . Students will engage in the complete engineering cycle, including conceptual design, mechanical fabrication, sensor integration, calibration, programming, and experimental validation. The system will measure load–deflection behavior during tensile tests and compute elastic modulus through real-time data processing. The project promotes hands-on learning in mechanics of materials, instrumentation, embedded systems, and digital manufacturing.	Dr Ravinder Kumar Duvedi	9780474711	rduvedi@thapar.edu	Dr Ravinder Kumar Duvedi, Dr Bikramjit Sharma	bikramjit@thapar.edu	9417871105	ELC
74	Design and Fabrication of a digital DIY Machine Kit for Fatigue Testing of Polymer samples.	This learning project focuses on the design and fabrication of a portable cyclic bending test machine integrated with a load cell and Arduino-based data acquisition system to evaluate the fatigue testing of polymer samples. Students will engage in the complete engineering cycle, including conceptual design, mechanical fabrication, sensor integration, calibration, programming, and experimental validation. The system will measure load–deflection behavior during fatigue strength through real-time data processing. The project promotes hands-on learning in mechanics of materials, instrumentation, embedded systems, and digital manufacturing.	Dr Sudhir Kumar		sudhir.kumar@thapar.edu	Dr Ravinder Kumar Duvedi, Dr Bikramjit Sharma	bikramjit@thapar.edu	9417871105	ELC
75	Autonomous Self-balancing robot	This project is to develop a self-balancing robot with improved design and control derived from its mathematical model. This robot not only equipped with IMU but also RGBD camera and lidar for autonomous navigation. For computation, it will be equipped with a Raspberry Pi 8gb for supporting ROS. As compared to the previous project, we will focus on mechanical aspect and for improved controller design.	Dr Raja Rout	9861014841	raja.rout@thapar.edu	Dr. Sunil Kumar			MED
76	Hybrid Model of TENG and EMG Flooring Tiles	Urban spaces are teeming with untapped kinetic energy from everyday human movement, yet most of this potential goes to waste. As cities strive for sustainability and smarter infrastructure, there is a pressing need for innovative solutions that can efficiently harvest and utilize this energy. Our project addresses this gap by developing a hybrid TENG-EMG kinetic flooring system, designed to capture energy from foot traffic in high-activity areas. By integrating advanced triboelectric and electromagnetic technologies, our approach promises greater efficiency, scalability, and real-world impact, paving the way for greener cities and smarter, self-powered public spaces.	Dr. Jayant Kolte	9960338240	jayantkolte@thapar.edu	Dr Ravinder Kumar Duvedi	rduvedi@thapar.edu	9780474711	ELC

77	Development of a Novel Controlled Diode Bridge Clamped (CDBC) Inverter with Advanced PWM Control	The experiential learning project aims to equip undergraduate students from Electrical Engineering, Instrumentation & Control Engineering, and Mechatronics Engineering with hands-on experience in modern inverter topologies and real-time control systems. The project focuses on the Controlled Diode Bridge Clamped (CDBC) inverter, a novel multi-level inverter architecture designed for improved efficiency and reduced switching losses compared to conventional T-Type VSI/ZSI inverters. A key aspect of this project is the implementation of advanced Pulse Width Modulation (PWM) control using Texas Instruments TMS320F28379D DSP, enabling precise switching and enhanced system performance. Students will gain exposure to power electronics design, inverter modeling, and embedded control systems, bridging the gap between theoretical learning and real-world applications. During the project, students will work on MATLAB/Simulink-based simulations to analyze inverter operation, develop optimized PWM switching techniques, and implement real-time pulse generation on the TMS320F28379D DSP. The hands-on activities include programming ePWM modules in Code Composer Studio (CCS), interfacing DSP with gate driver circuits, and validating inverter performance through experimental testing. This approach enables students to understand efficiency optimization, harmonic distortion reduction, and practical challenges in high-power inverter applications. By the end of the project, students will have designed, simulated, and tested an advanced inverter control system, gaining valuable skills in embedded programming, digital control techniques, and industrial power electronics. The developed inverter is expected to have a significant impact in renewable energy systems, electric vehicles, and industrial motor drives, where high-efficiency power conversion is essential. With DSP-based real-time control, the proposed topology enhances energy conversion efficiency, reduces size, and improves performance stability. The project provides students with a strong technical foundation in power electronics and embedded control, preparing them for careers in renewable energy, electric mobility, and automation industries. By integrating simulation, hardware implementation, and performance analysis, this initiative fosters innovation and practical learning.	Dr Santosh Sonar	9933135606	santosh.sonar@thapar.edu	Dr Abhisek Pal	abhisek.pal@thapar.edu	89005 15980	EIED
78	HarmonyHub: Design and Development of Music Synthesizer	Pilots for New Experiential Activities	Dr. Sangita Roy	8002396506	sangita.roy@thapar.edu	Dr. Sharad Saxena	sharad.saxena@thapar.edu	8195098711	CSED